

Investigating the Molecular Mechanisms of Prostate Cancer Bone Metastasis

Ella Yee, Florencia Picech, Cory Abate-Shen
Department of Molecular Pharmacology and Therapeutics, Columbia University



Background

- Metastasis is the leading cause of prostate cancer death
- Bone is the most common site of prostate cancer metastasis
- Current therapies for bone metastasis are not curative
- Molecular mechanisms of bone metastasis are poorly understood
 - Androgen receptor (AR) signaling

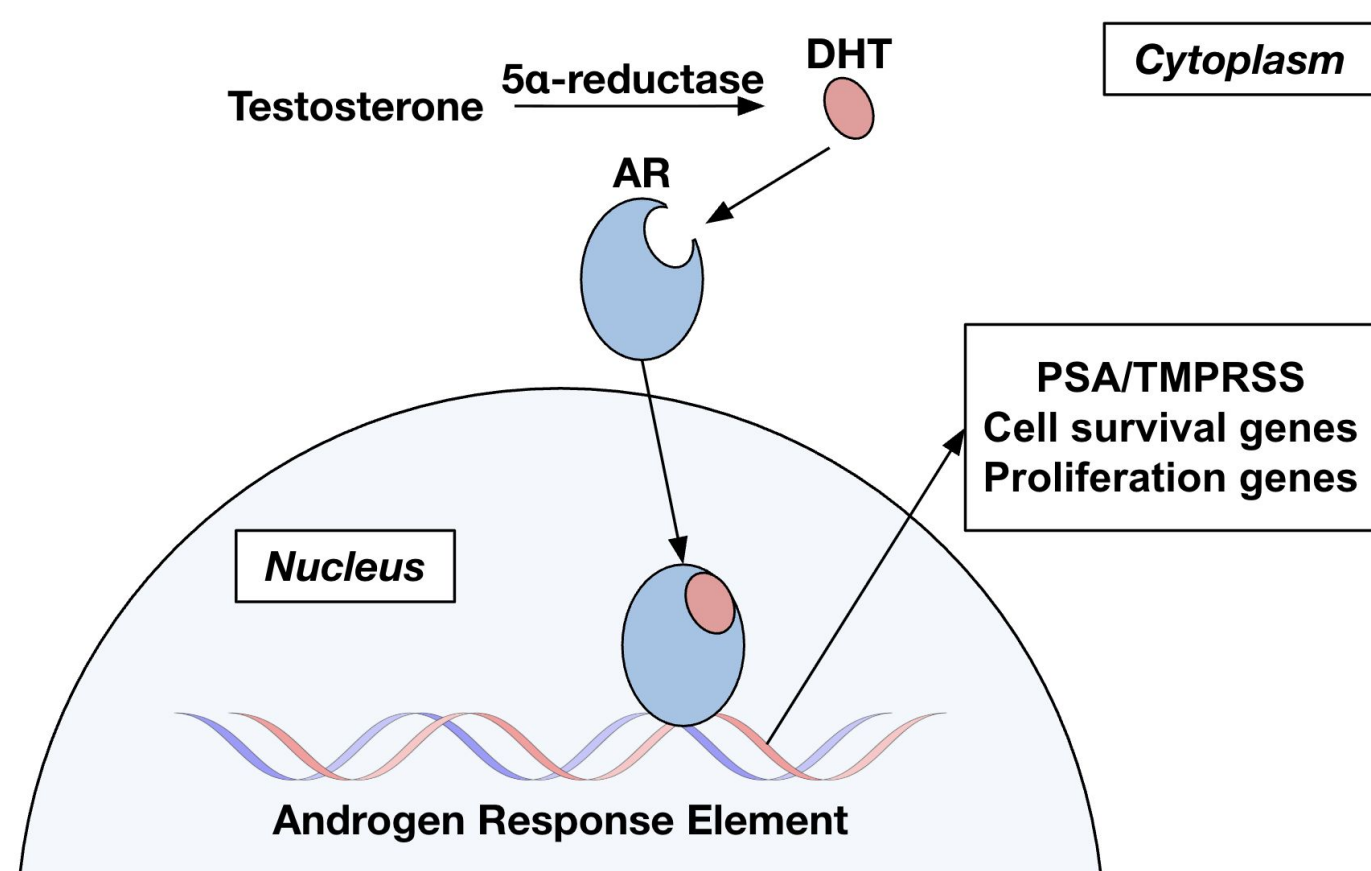


Figure 1. Illustration of androgen receptor signaling. Testosterone is converted to DHT, which binds AR and promotes nuclear translocation and activation of androgen-responsive genes involved in proliferation and survival.

- Epithelial-mesenchymal transitions (EMT)

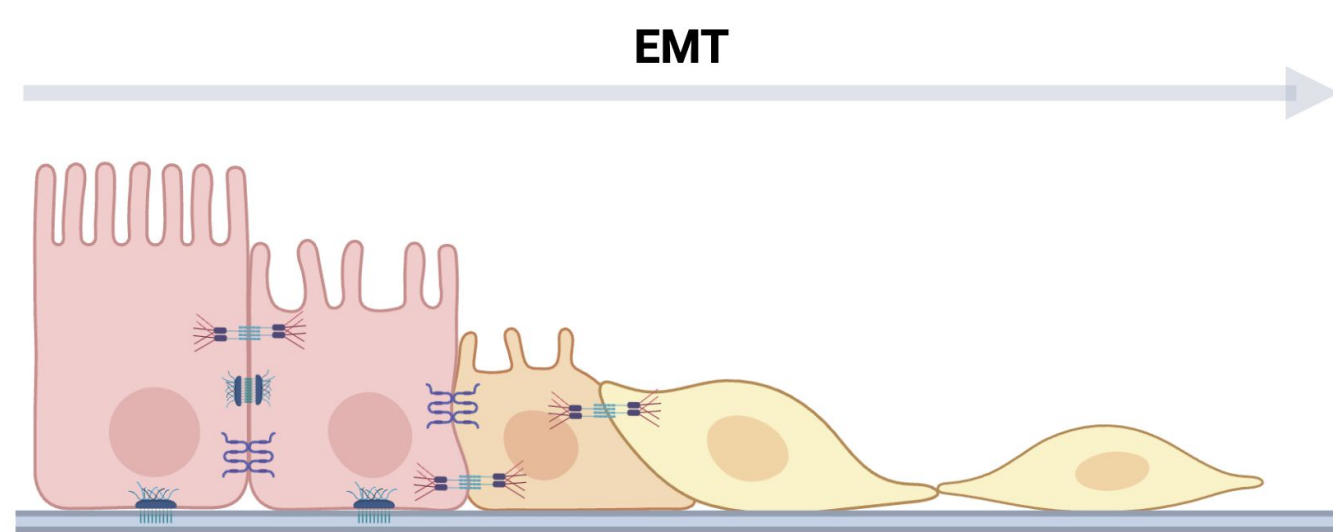


Figure 2. Illustration of epithelial-mesenchymal transition. The process by which epithelial cells lose cell-cell adhesion and develop mesenchymal features is depicted from left to right.

Research Questions

- Do distinct molecular mechanisms drive prostate cancer bone metastasis?
 - Comparison across prostate cancer subtypes
- Do bone metastases retain the molecular identity of the primary tumor?

Methods

Genetically-Engineered Mouse Models (GEMMs)

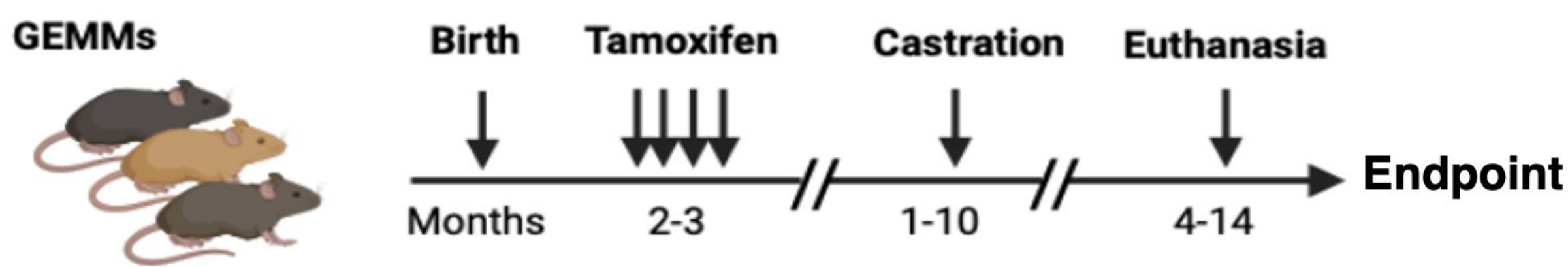


Figure 3. Development of genetically-engineered mouse models. Mice are administered tamoxifen at 2-3 months of age, monitored throughout disease progression, and euthanized by carbon dioxide inhalation at the clinical endpoint.

Table 1. Genetically-engineered mouse model genotypes. GEMMs recapitulate different phenotypes of human prostate cancer from pre-invasive to metastatic and allow for the study of cancer progression in the context of the whole organism.

GEMM	Genotype
NP (Control)	Nkx3.1 ^{CreERT2/+} ; Pten ^{flox/flox} ; R26R-CAG-LSL-EYFP/+
NPp53Rb1 ^{+/-} (Experimental)	Nkx3.1 ^{CreERT2/+} ; Rb1 ^{flox/+} ; Pten ^{flox/flox} ; p53 ^{flox/flox} ; R26R-CAG-LSL-EYFP/+
NPK (Experimental)	Nkx3.1 ^{CreERT2/+} ; Pten ^{flox/flox} ; Kras ^{LSL-G12D/+} ; R26R-CAG-LSL-EYFP/+

Methods

Genotype Comparison

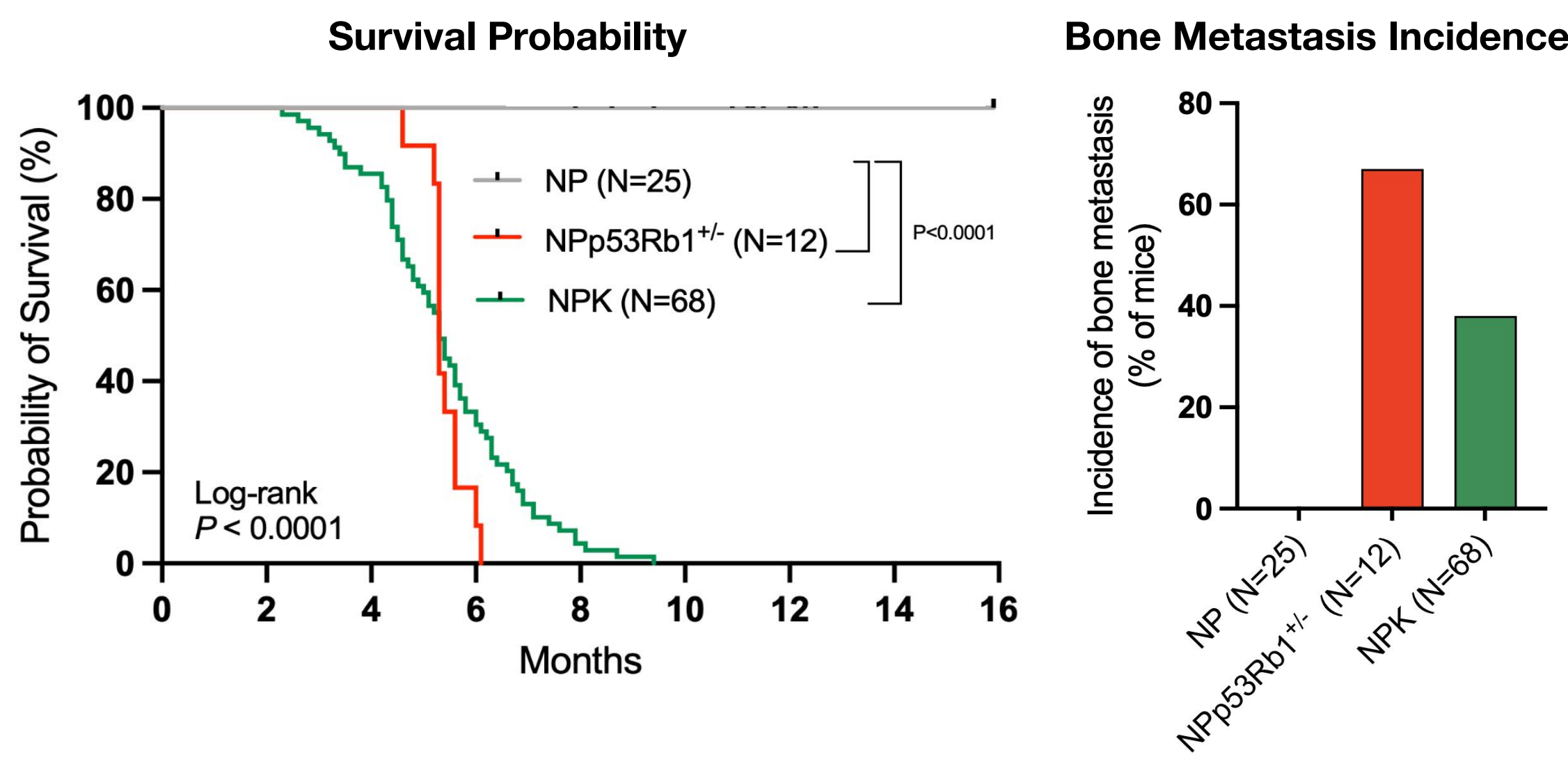


Figure 4. Survival probability and bone metastasis incidence of GEMMs. NPp53Rb1^{+/-} and NPK mice were selected as experimental genotypes due to long survival and high bone metastasis incidence.

GEMMs Analysis Approach

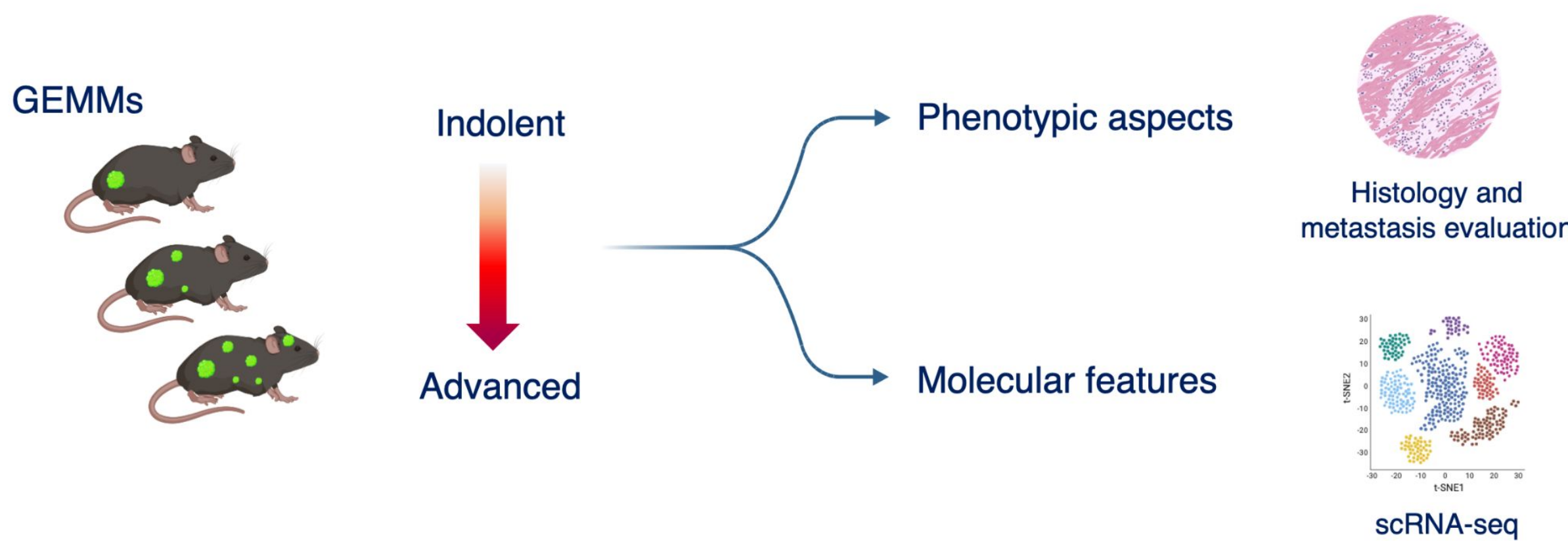


Figure 5. Analysis pipeline for genetically-engineered mouse models. This project uses immunohistological analysis of GEMMs to better understand phenotypic aspects of prostate cancer bone metastasis.

Experimental Workflow

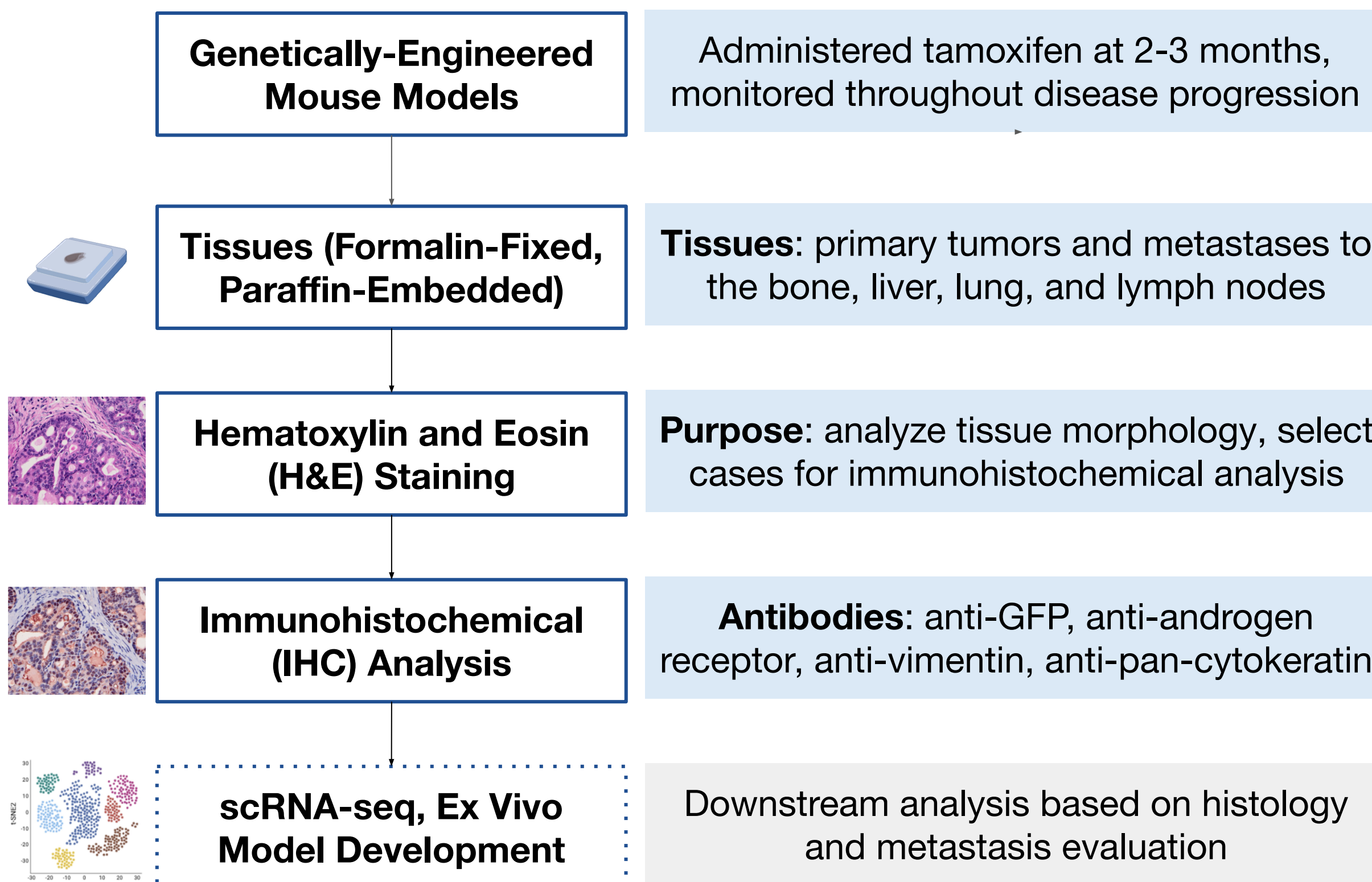


Figure 6. Experimental workflow. Tissues were collected from genetically-engineered mouse models of three genotypes. Selected cases were studied via hematoxylin and eosin staining and immunohistochemical analysis.

IHC Primary Antibodies:

- Anti-GFP: identify primary tumors, metastatic lesions
- Anti-androgen receptor (AR): study androgen receptor signalling
- Anti-pan-cytokeratin (PanCK): mark epithelial, luminal cells
- Anti-vimentin (VIM): mark mesenchymal cells

Results

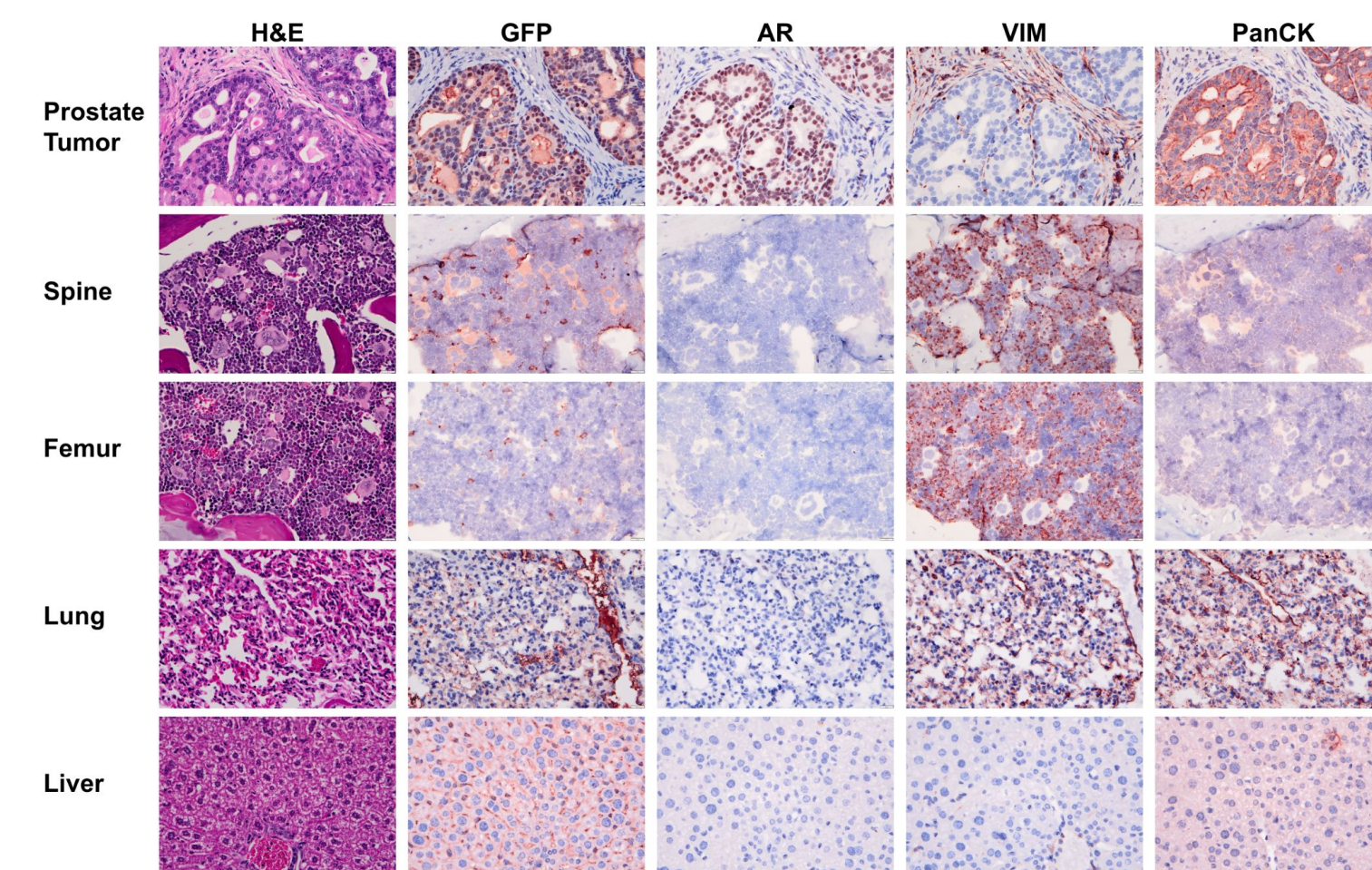


Figure 7. Histopathological analysis of NP mice. Primary tumors of NP mice expressed AR and PanCK. Images of the spine, femur, lung, and liver were taken from normal tissue. These sections were negative for AR and PanCK. VIM was expressed in bone marrow.

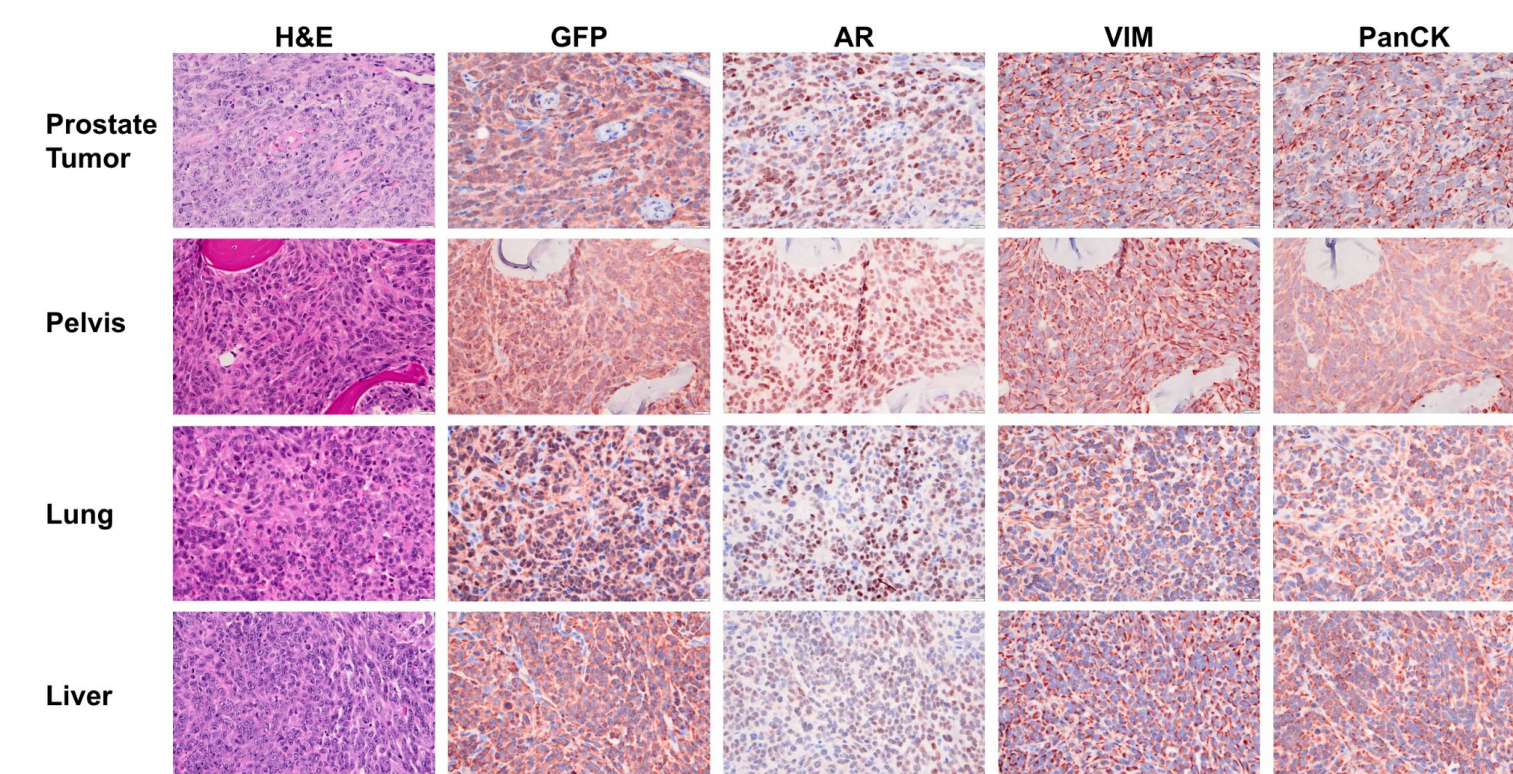


Figure 8. Histopathological analysis of NPK mice. Primary tumors of NPK mice were positive for AR, VIM, and PanCK. Most metastatic lesions were positive for AR, VIM, and PanCK.

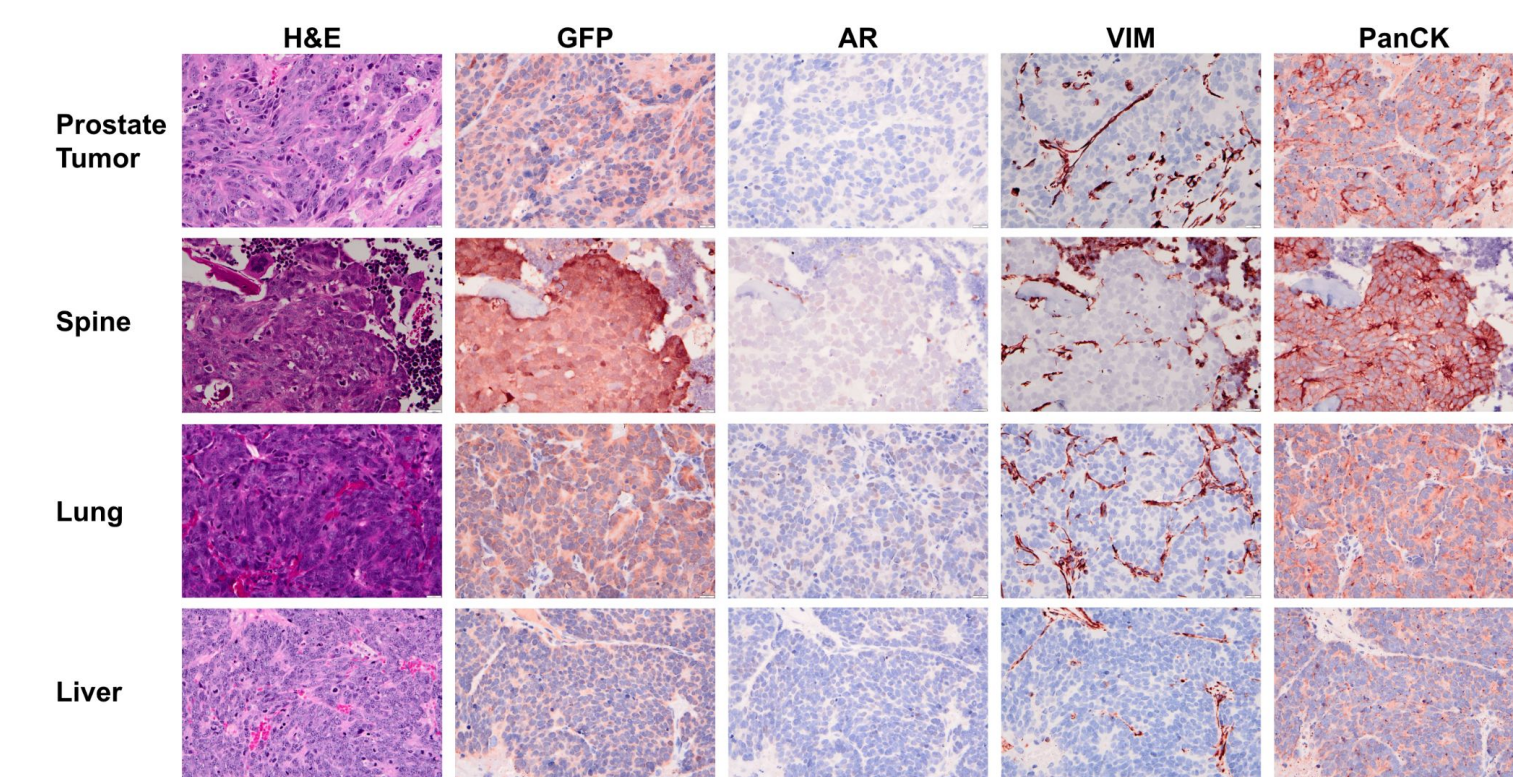


Figure 9. Histopathological analysis of NPrb1^{+/-} mice. Primary tumors of NPrb1^{+/-} mice were positive for only PanCK. All metastatic lesions were also only positive for PanCK.

Table 2. Summary of immunohistochemistry. Metastatic lesions retained the characteristics of their primary tumors in both experimental genotypes, while staining for AR and VIM differed.

Genotype	Androgen Receptor (AR)				Vimentin (VIM)				Pan-Cytokeratin (PanCK)			
	Primary	Lung	Liver	Bone	Primary	Lung	Liver	Bone	Primary	Lung	Liver	Bone
NP	+	N/A	N/A	N/A	-	N/A	N/A	N/A	+	N/A	N/A	N/A
NPK	+	+	+	+	+	+	+	+	+	+	+	+
NPp53Rb1(+/-)	-	-	-	-	-	-	-	-	+	+	low	+

Discussion & Conclusions

- Expression in metastatic lesions corresponded to those of primary tumors**
 - Primary tumors retain their molecular phenotype during metastatic progression
- Genotypes reflect subtype-specific mechanisms of bone metastasis**
 - NPK mice express androgen receptor (AR), vimentin (VIM), and cytokeratins (PanCK) in all tissues
 - NPp53Rb1^{+/-} mice expressed only cytokeratins (PanCK) in all tissues
- Future directions:**
 - Expand sample size
 - Investigate androgen deprivation in GEMMs

Acknowledgements

- Thank you to Dr. Florencia Picech for sharing experimental data and providing mentorship throughout this project.
- Thank you to Dr. Cory Abate-Shen, as well as the Egleston scholars and SURF programs, for the research opportunity.